Artist-Name Responses beyond a Shared Painting Effect in Text-to-Image Generation

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Abstract

A detectable response to a painter's name need not reproduce the differences between that painter's works and those of other painters. We examine this distinction in 31 interpretable digital features using 649 reproductions of Monet, Sisley, Pissarro and Cézanne. Six generators receive 14 fixed outdoor briefs under artist-free, generic-painting and four named-painter conditions, with two outputs per cell. Removing the common additive response separates reference-aligned contrast strength from repeat-corrected disagreement with pooled reference contrasts. All six configurations have positive aligned slopes. GPT Image 2 has a slope close to one ($\beta = .999$), but substantial contrast error ($D = 1.226$). FLUX.2 Max has the lowest estimated primary error and adjusted advantages over the two GPT Image 2.5 configurations; no model is established to outperform the no-artist-contrast benchmark. Post-result diagnostics qualify this ordering: scalar calibration evaluated on held-out scenes favors GPT Image 2, and genuine-painting controls expose substantial content dependence and reference uncertainty. The study distinguishes detectable name effects, response calibration and agreement with a specified reference collection. It does not validate this feature-space agreement as artistic fidelity.

1 Introduction

A painter's name can change an image in several ways. It can introduce a common painting appearance, exaggerate familiar color or texture cues, or alter the features that distinguish one painter from another. These effects are easy to conflate. Moving closer to Monet's collection, for example, does not show that a generator reproduces the differences between Monet and Sisley: a generic painting instruction could improve proximity to both.

We ask whether painter names induce differences aligned with those among *documented digital reproductions*, beyond a response shared across names. Our target is the labeled, centered mean configuration of four painters in 31 color, spatial and texture coordinates. This is an explicit collection-relative target, not a perceptually validated definition of style. Historical subject mixtures and reproduction processes can contribute to its artist differences. Holding the generated scene text fixed controls the prompt intervention; it does not make those historical subjects identical.

The distinction matters for evaluation. A generator can produce easily separable named conditions whose labels map to the wrong reference painters. It can also recover the reference-aligned amplitude while introducing large additional contrasts. Conversely, a restrained response may have lower squared error than a stronger response, without distinguishing artists more accurately in every respect. A single proximity score or recognition accuracy cannot separate these possibilities.

We cross six model configurations with identical scene descriptions and artist-free, generic oil-painting and four named-painter clauses. Centering the named outputs removes their common additive shift. A reference-aligned slope measures response amplitude; an independent-repeat cross-product estimates scene-wise disagreement with the pooled reference contrasts without additive sampling-noise bias. Paired scene comparisons retain the same target across models.

This applies established estimation principles to a controlled audit of low-level artist contrasts, rather than proposing a new general style metric.

The experiment supports positive reference alignment but not a general ranking of artistic fidelity. Review-driven diagnostics reveal two reasons for that restriction. First, scene-wise disagreement includes variation in artist contrasts across scenes, which can be appropriate rather than erroneous. Second, response magnitude influences the primary model ordering: a scalar fitted on other scenes changes that ordering. We report these post-result checks separately from the declared analysis, alongside artist-pair summaries, genuine-painting controls and inspectable image examples. They explain aspects of the measured disagreement while leaving its perceptual and causal interpretation open.

2 Related work

Artist imitation and computational style measurement are established research problems. Somepalli et al. (2024) develop learned style descriptors and study style similarity across diffusion models. Su et al. (2025) evaluate prompted-artist recognition using content-controlled name substitutions; they also compare named and artist-free images with real-artist prototypes in CLIP space. Their work already establishes that artist names can leave recognizable signatures. Our question is whether those signatures recover the *relative geometry of the reference painters*, after removing the shared response. Neither name substitution nor cross-model comparison is claimed as a new method here.

Deliège et al. (2025) examine stylistic imitation and stereotyping with expert assessments of historical and generated art. AI-Pastiche studies authenticity judgments and prompt adherence (Asperti et al., 2025); AI-WikiArt uses painting-derived captions and vision-language models to study attribution (Fu et al., 2025). These studies motivate distinguishing convincing imitation, recognizable authorship and distributional recovery. We test reference-relative painter differences and summarize full-distribution outcomes in explicit digital coordinates, without substituting them for expert judgments.

Quantitative art research has examined color, composition and image statistics across painting collections (Kim et al., 2014; Sigaki et al., 2018; Lee et al., 2020). Kim et al. (2026) combine formal and contextual representations to study art-historical change, including contextual image-to-image experiments. Our intervention instead adds painter names to text-only scene prompts, with no original composition supplied to the generator. Related content and style remain entangled in our reference panel; the common-scene intervention controls the requested content of generated images, not the content of historical works.

Generative-model evaluation distinguishes fidelity and diversity (Naeem et al., 2020), and conditional metrics distinguish variation within conditions from relationships among their means (Benny et al., 2021). Our cross-repeat estimator follows the established principle of estimating squared signal differences from independent partitions (Diedrichsen and Kriegeskorte, 2017). Energy statistics provide the complementary distributional endpoint (Székely and Rizzo, 2013). Processing can materially change image comparisons (Parmar et al., 2022); separation in a learned embedding is likewise not itself perceptual validation (Asperti, 2026). Together, these precedents motivate a test of reference-relative artist contrasts with generic controls and independent repeats. The contribution is the combined experimental test and its findings, rather than a new style metric.

3 Design and measurements

3.1 Painters and reference images

The primary panel comprises 649 distinct painting reproductions: 297 Monet, 106 Sisley, 141 Pissarro and 105 Cézanne. Artist means receive equal weight, so the larger Monet collection does not dominate the between-artist target. A separate development panel of 221 works supplies feature scaling; no primary reference or generated image is used to fit that transformation. Both panels were assembled before the present experiment and had been used in earlier analyses. They are explicit, finite comparison panels rather than newly withheld samples of each painter's complete oeuvre.

Reference files are linked to recorded work identities and Wikimedia Commons source metadata. Commons supports traceable file histories, source and rights fields, and institutional collection contributions through its cultural-heritage partnerships (Wikimedia Commons, 2026). Wikidata supplies structured, reusable work and artist identities (Vrandečić and Krötzsch, 2014). These properties make the selection and provenance inspectable; they do not guarantee uniform photography, color calibration or conservation state. We therefore call the inputs *digital reproductions*, not direct measurements of physical paintings. Previous computational art corpora support the use of documented reproductions for comparative analysis, but do not authenticate this collection: the Web Gallery of Art in Kim et al. (2014), ART500K in Kim et al. (2026), and WikiArt in Fu et al. (2025) are distinct from Wikimedia Commons.

3.2 A common-scene, six-model experiment

We compare GPT Image 1, GPT Image 2, GPT Image 2.5 Flare, GPT Image 2.5 Sunburst, Nano Banana 2 and FLUX.2 Max. Fourteen authored outdoor scenes span water, built, land and mixed compositions. Each scene is crossed with six clauses: no painting instruction; "Render as an oil painting"; and "Render as an oil painting in the style of [painter]" for each of the four painters. The scene text is otherwise identical. Each model–scene–clause cell receives two separately requested outputs, giving 1,008 images collected on 10 September 2026.

All models receive text only and return 1,024-by-1,024-pixel images. The four OpenAI models use medium quality; the other two use their specified 1K configurations. These comparisons concern the recorded configurations, not maximum attainable quality or equal expenditure per image. The exact prompts, parameters and randomized request order were fixed before collection. The 14 scenes are a fixed design, not a random sample of all possible content.

3.3 What the features measure

Images undergo orientation correction and conversion to sRGB using valid embedded profiles; compatible unprofiled files are treated as sRGB. The primary pipeline preserves aspect ratio at a 512-pixel short side, using Lanczos resizing without upsampling. No painting-region mask is applied: source margins or calibration strips can enter the measurements. Each coordinate is centered and scaled by its median and interquartile range in the development panel, with equal total weight per painter. Thus a unit is a development-panel IQR in that coordinate, not a perceptual just-noticeable difference. The Euclidean metric weights standardized coordinates equally, without further equalizing feature families.

Table 1 explains the representation. Color features summarize lightness and chroma, concentration of hue, and color changes across spatial scales; CIEDE2000 supplies the color-difference calculation (Sharma et al., 2005). Spatial features describe how luminance contrast is organized across frequencies, directions and regions. Texture features summarize multiscale detail and local intensity patterns using wavelets and local binary patterns (Mallat, 1989; Ojala et al., 2002). For example, similar mean chroma can coexist with different hue diversity, and similar total contrast can coexist with different edge orientation or fine-scale texture. These distinctions make family-specific results interpretable, while correlated coordinates and processing sensitivity prevent treating the Euclidean metric as a validated measure of artistic style.

4 Measuring reference-relative artist contrasts

4.1 Reference-aligned variation

Let $\mu_a \in \mathbb{R}^{31}$ be the standardized reference mean for painter a and let $\bar{\mu} = \frac{1}{4}\sum_a \mu_a$. Define the reference painter contrasts $r_a = \mu_a - \bar{\mu}$ and their total squared magnitude $H = \sum_a \|r_a\|^2$. For generated vector z_{msak} from model m, scene s, painter a and repeat $k \in \{1, 2\}$, remove the common additive named response:

$$d_{msak} = z_{msak} - \frac{1}{4}\sum_b z_{msbk}. \tag{1}$$

Any shift shared by all four named outputs of that model, scene and repeat cancels exactly. This removes additive baselines in these coordinates. It does not remove a shared transformation: if all conditions undergo $z \mapsto Az + b$, centering removes b but leaves Ad.

The reference-aligned slope is

$$\widehat{\beta}_{msk} = \frac{\sum_a \langle d_{msak}, r_a \rangle}{H}, \qquad \widehat{\beta}_m = \frac{1}{S}\sum_s \frac{\widehat{\beta}_{ms1} + \widehat{\beta}_{ms2}}{2}. \tag{2}$$

It is the least-squares coefficient of generated painter contrasts on reference contrasts. A value of zero indicates no response along the reference direction; one matches its amplitude. Values above one indicate exaggeration along that direction, not necessarily better recovery. A nonzero slope does not establish agreement for every artist pair. It is a projection onto one vectorized reference configuration, not a test that all reference contrast dimensions are recovered.

Family	Measurements	Interpretation
Color (11)	Lightness and chroma median/IQR; chromatic fraction; hue concentration and entropy; color differences at three lags and their slope	Brightness, color strength and diversity, and how rapidly colors change over space.
Spatial (8)	Spectral slope, residual and anisotropy; orientation entropy; horizontal–vertical balance; gradient median/IQR; quadrant divergence	Coarse versus fine organization, dominant directions, edge strength and regional variation in orientation.
Texture (12)	Four wavelet energies, slope and curvature; three local-binary-pattern entropies; three local coefficients of variation	Detail across scales, diversity of local patterns and local luminance contrast.

Table 1: The 31 fixed digital features. Counts refer to scalar coordinates. Texture measurements do not measure physical paint relief or authenticate brushwork. Exact scales and definitions appear in Appendix A.

4.2 Repeat-corrected disagreement with a pooled target

Two independently requested repeats permit the cross-product estimator

$$\widehat{D}_{ms} = \frac{1}{H}\sum_a \langle d_{msa1} - r_a,\ d_{msa2} - r_a\rangle, \qquad \widehat{D}_m = \frac{1}{S}\sum_s \widehat{D}_{ms}. \tag{3}$$

Under stable conditional means and independent, mean-zero repeat errors, its expectation is the squared error of the scene-conditional mean painter contrasts, normalized by H. The error contributed by output sampling does not inflate this expectation. Exact agreement with these reference contrasts in every scene gives expected $D = 0$; identical conditional means for all four artist labels give expected $D = 1$. This latter benchmark is a theoretical no-contrast predictor, not an observed generic-arm estimate. Finite estimates may be negative. We do not truncate them or interpret a negative estimate as better-than-perfect recovery.

The estimator has an exact decomposition. Write $e_{msak} = d_{msak} - \widehat{\beta}_{msk} r_a$. Then

$$\widehat{D}_{ms} = (\widehat{\beta}_{ms1} - 1)(\widehat{\beta}_{ms2} - 1) + \frac{1}{H}\sum_a \langle e_{msa1}, e_{msa2}\rangle. \tag{4}$$

The first term measures reference-aligned amplitude error. The residual is orthogonal to the single stacked reference configuration; it is not necessarily outside the feature-space span of the reference means, much less outside all authentic artistic variation. Both terms can be negative in finite samples.

A simple example clarifies the target. Conditional contrasts 0, r and $2r$ give expected errors 1, 0 and 1, respectively. Permuting distinct reference labels preserves all unlabeled pairwise distances but gives nonzero error. Thus D measures labeled coordinate agreement, including response strength. It is not just a measure of separation or distance-geometry preservation.

Let θ_s stack the four conditional mean contrasts and $\bar{\theta} = S^{-1}\sum_s \theta_s$. The population target decomposes as

$$D_{\text{scene}} = \frac{\|\bar{\theta} - r\|^2}{H} + \frac{1}{SH}\sum_s \|\theta_s - \bar{\theta}\|^2 = D_{\text{aggregate}} + V_{\text{scene}}. \tag{5}$$

The second term penalizes changes in artist contrasts across scenes. Such changes need not be artistic mistakes: a generator with $\theta_s = r + u_s$ and $\sum_s u_s = 0$ matches the aggregate target but incurs positive scene-wise error. We therefore interpret the primary endpoint as *scene-wise agreement with pooled reference contrasts*. It cannot establish content-conditional artist fidelity without a corresponding historical target.

4.3 Comparing models

We fit the scene fixed-effects regression

$$\widehat{D}_{ms} = \alpha_s + \gamma_m + \varepsilon_{ms}, \qquad \gamma_{\text{GPT Image 2}} = 0. \tag{6}$$

All models face the same scenes and equally weighted reference painter target. With a complete balanced panel, model coefficient differences equal the mean paired scene differences. Inference uses those scene differences rather than treating features, painters or individual images as independent replicates.

The fixed inferential family contains six reference-aligned slopes and 15 pairwise model error contrasts. We report paired-scene Student intervals with $n-1$ degrees of freedom and Bonferroni adjustment across these 21 comparisons. Nominal 95% intervals for absolute D_m are descriptive. These are approximate model-based summaries using dispersion across the fixed briefs. That dispersion mixes generation noise with genuine between-brief variation in conditional contrasts; it does not isolate fresh-request uncertainty for exactly these briefs or establish coverage for new scenes. Bonferroni adjustment requires adequately calibrated individual intervals. Two outputs permit noise correction but do not precisely characterize its sampling distribution. All 14 scenes are complete for all models.

Prespecified sensitivity analyses remove each scene in turn, resample scenes for model contrasts, and resample reference works within painter while holding generated vectors fixed. The last analysis assesses dependence on this reference panel; it is not a second independent uncertainty estimate. We also repeat the analysis in each feature family, without texture, and using central square measurement windows for both generated and reference images. The square view controls measurement-window aspect ratio while discarding peripheral content. These analyses are descriptive and do not select a preferred representation.

A separately declared sensitivity addresses unequal subject mixtures in the references. Before inspecting the new feature outcomes, we specified equal weight for four previously recorded, metadata-derived content classes within each painter, then recomputed the reference contrasts and all model estimates. For example, water scenes are much more frequent in the Monet panel than in the Pissarro panel. Equal class weighting changes the pooled mixture; it does not supply a content-conditional target, match individual compositions or identify pure style. Reference resampling within artist/class strata assesses the stability of this alternative target.

4.4 Generic controls and full distributions

The mean change from the free arm to the four named arms separates into a common component and centered painter contrasts. We compare the direction and magnitude of the common component with the free-to-generic change. Their cosine similarity and cross-repeat squared magnitudes test whether a generic painting clause reproduces the shared direction. A large common fraction alone cannot show that naming fails: these four reference painters also share visual properties. Equations 2–3 separately assess reference contrasts.

For painter a, energy discrepancy compares its generated collection Y_a with reference collection X_a:

$$\widehat{\mathcal{E}}(X,Y) = \frac{2}{n_X n_Y}\sum_{i,j}\|x_i - y_j\| - \frac{1}{n_X^2}\sum_{i,i'}\|x_i - x_{i'}\| - \frac{1}{n_Y^2}\sum_{j,j'}\|y_j - y_{j'}\|. \tag{7}$$

Unlike distance to a mean, this statistic incorporates within-collection variation. Its finite-sample expectation depends on within-collection spread, so equal generated sample sizes do not eliminate comparison bias. We retain this declared statistic and add a correction respecting the scene-stratified design in Appendix C. Generated/reference variance ratios compare 28 outputs from 14 fixed briefs with a historically broader collection. A ratio below one is not, by itself, evidence of deficient diversity. We also report within-scene repeat spread, $\frac{1}{2S}\sum_s \|z_{msa1} - z_{msa2}\|^2$, the mean sample-variance trace of the two repeated outputs. This separates repeat dispersion from total variation across briefs.

4.5 Post-result diagnostic analyses

The declared endpoints are retained. The following analyses were added in response to review, after the primary results were known, and are descriptive. We examine artist pairs, reference singular components, label permutations, class-specific targets and disjoint real/reference controls. Equal family weighting and a covariance metric fitted only to development works examine coordinate weighting; simulation probes the interval procedure under explicit noise models. Appendix C specifies these choices.

To separate contrast strength from direction, define

$$\widehat{Q}_m = \frac{1}{SH} \sum_{s,a} \langle d_{msa1}, d_{msa2} \rangle, \qquad \widehat{D}_m = 1 - 2\widehat{\beta}_m + \widehat{Q}_m. \tag{8}$$

For positive $\widehat{Q}_m$, $\widehat{\beta}_m/\sqrt{\widehat{Q}_m}$ is a noise-corrected alignment summary; finite estimates need not lie in $[-1, 1]$. Scaling every centered generated contrast by c gives $\widehat{D}_m(c) = 1 - 2c\widehat{\beta}_m + c^2\widehat{Q}_m$. We fit $c = \max(0, \widehat{\beta}/\widehat{Q})$ on the other 13 scenes and evaluate it on the held-out scene. The fitting criterion is available only when its training $\widehat{Q}$ is positive, as it is in every observed fold. This counterfactual concerns feature vectors, not an implemented image transformation.

5 Results

5.1 Reference-aligned responses do not establish low absolute error

All 1,008 outputs yield valid measurements. Every aggregate reference-aligned slope has a simultaneous interval above zero (Table 2). This excludes a purely common additive response under the stated inference model. It does not establish agreement for every artist pair or low total error. GPT Image 2 has a slope close to one ($\beta = .999$, simultaneous interval $[.893, 1.104]$), while $D = 1.226$. Its orthogonal residual contributes 1.229; the corrected amplitude-error term is $-.003$. No amplitude-equivalence test was specified.

FLUX.2 Max has the smallest estimated primary error, .801, with a weaker aligned slope, .470. The fixed 21-comparison procedure resolves Flare minus FLUX at .937 $[.256, 1.618]$ and Sunburst minus FLUX at .840 $[.058, 1.623]$. The other 13 pairwise intervals include zero. These comparisons favor FLUX on the declared endpoint and scene panel; they do not identify a uniquely best generator. Five of six error point estimates exceed the no-contrast benchmark of one, and even FLUX's nominal interval, $[.586, 1.016]$, includes one. No configuration is established to beat that benchmark. Absolute-error intervals were not part of the fixed inferential family.

Figure 1 illustrates the generated contrasts and their labeled reference target. These reference-only axes display 86.99% of reference-centroid variation; all numerical comparisons use 31 coordinates.

Model	Reference slope β	Error D	ΔD versus GPT Image 2
GPT Image 1	0.938 $[0.709, 1.168]$	1.572	0.346 $[-0.400, 1.093]$
GPT Image 2	0.999 $[0.893, 1.104]$	1.226	0 (baseline)
GPT Image 2.5 Flare	0.772 $[0.680, 0.864]$	1.738	0.512 $[-0.126, 1.150]$
GPT Image 2.5 Sunburst	0.824 $[0.680, 0.968]$	1.641	0.415 $[-0.350, 1.181]$
Nano Banana 2	0.442 $[0.256, 0.629]$	1.109	-0.116 $[-1.155, 0.922]$
FLUX.2 Max	0.470 $[0.239, 0.701]$	0.801	-0.425 $[-1.011, 0.161]$

Table 2: Pooled-reference agreement and scene-paired model comparisons. Lower D indicates less scene-wise mismatch with pooled reference contrasts; larger β is not necessarily better. Brackets give approximate simultaneous intervals from the fixed family of six slopes and 15 model contrasts. On a complete common panel, model differences equal the scene fixed-effects regression contrasts. Inference uses paired scene differences.

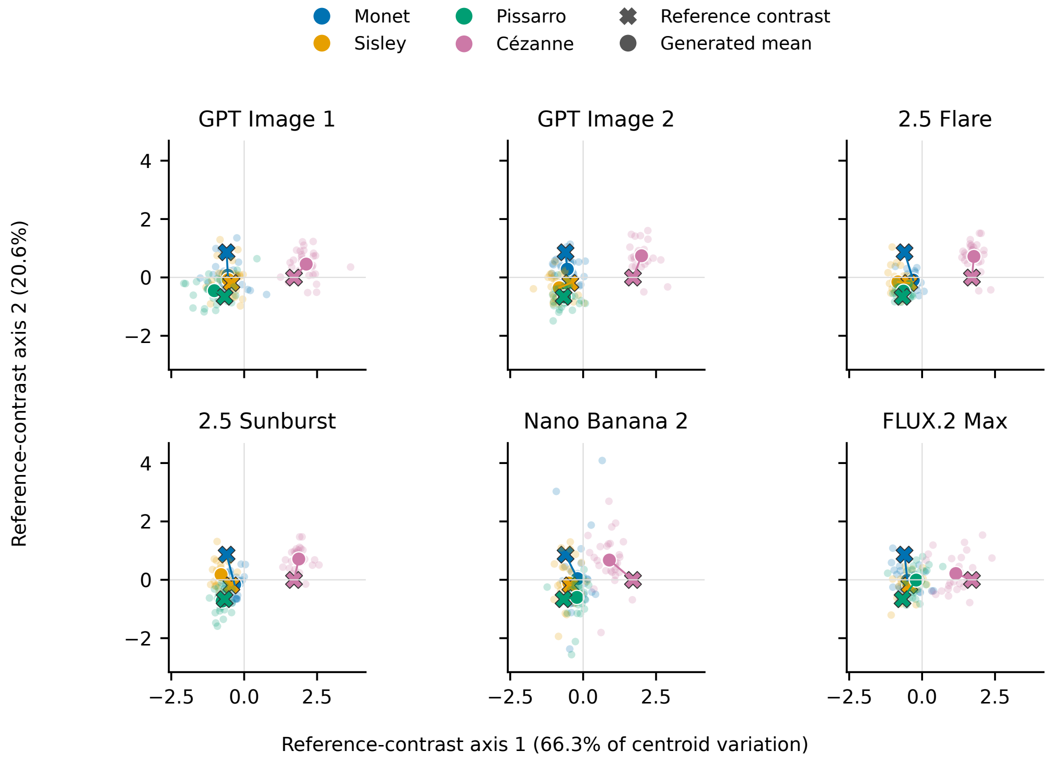

Figure 1: Centered painter contrasts. Colors identify painters; faint points are generated contrasts, circles their means, and crosses the reference contrasts. All panels share axes fitted to the four reference centroids. Separation of named conditions does not establish agreement with their labeled reference configuration or validate that configuration as a measure of artistic style.

5.2 Aggregate alignment masks uneven artist distinctions

Post-result artist-pair diagnostics show why the six positive slopes should not be interpreted as recovery of every painter distinction (Figure 2). Monet–Sisley slopes are .074 for GPT Image 1, −.011 for Flare, −.249 for Sunburst, −.044 for Nano Banana 2 and .129 for FLUX; GPT Image 2 has a larger estimate, .402. These are descriptive projections, not additional artist-level rejection tests.

The three nonzero reference components account for 66.3%, 20.6% and 13.0% of centroid variation. Every model has a larger slope on the dominant component than on either remaining component. Omitting Cézanne reduces FLUX's aggregate slope from .470 to .096; the corresponding GPT Image 2 slope changes from .999 to .651. The primary response is therefore not equally distributed across artist distinctions. Across all 24 label assignments, the correct assignment minimizes error for GPT Image 1, GPT Image 2 and FLUX, but ranks second for Flare, Sunburst and Nano Banana 2. For each of the latter three, swapping Monet and Sisley lowers the point estimate. These permutation ranks are descriptive; no label-exchangeability test is claimed.

5.3 Scene dependence and response magnitude qualify model rankings

Table 3 separates aggregate mismatch from scene variation. Nano Banana 2 has lower aggregate error than FLUX (.723 versus .761), but a larger scene-variation component (.387 versus .040), reversing the primary point ordering. This is a difference between evaluation targets; it is not evidence that all canceled scene variation is inappropriate.

Response magnitude also matters. GPT Image 2 has the highest corrected alignment ratio, .670, but generated contrast magnitude $Q = 2.223$ relative to the reference. FLUX has smaller

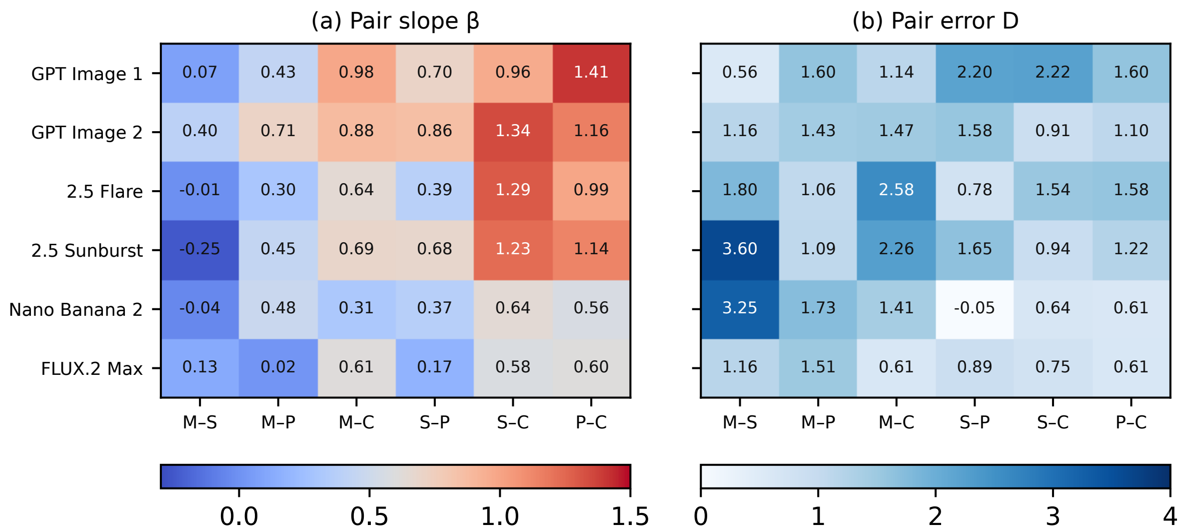

Figure 2: All six artist-pair contrasts, assessed after the primary results. M, S, P and C denote Monet, Sisley, Pissarro and Cézanne. Each pair uses its own reference difference: slope 1 matches its aligned amplitude, and expected error 1 denotes no difference between the two named conditions. Slopes and errors use both repeats and all 31 features. Positive model-level alignment can coexist with weak or reversed alignment for individual pairs.

magnitude, .741, and lower alignment, .546. A nonnegative scalar fitted on 13 scenes and evaluated on the omitted scene reduces mean error for every model. The smallest calibrated point estimate is GPT Image 2's .554, compared with FLUX's .714. This held-scene feature counterfactual supports a calibration explanation of the primary ordering, without changing the declared model comparison or establishing that such transformed features can be generated as images.

The historical target is itself content dependent. Contrasts among the four reference title classes differ from the pooled reference configuration by an average squared amount equal to $.438H$. On the 11 unambiguously designated water, built and land briefs, FLUX retains the lowest point estimate with a class-specific target (Appendix C). These broad labels do not match compositions or verify generated scene adherence.

Genuine-painting controls expose the limits of an absolute interpretation. Across 1,000 disjoint reference half-splits, two independently sampled works per painter and pseudo-scene give median error .233 for pooled sampling; the central 95% of simulated estimates span $[-.758, 1.380]$. Class-stratified sampling against pooled training targets gives median .721 and range $[-.554, 2.400]$.

Model	$D_{\text{aggregate}}$	V_{scene}	Q	$\beta/\sqrt{Q}$	D_{held}
GPT Image 1	1.239	0.333	2.449	0.600	0.645
GPT Image 2	1.044	0.182	2.223	0.670	0.554
2.5 Flare	1.483	0.255	2.281	0.511	0.741
2.5 Sunburst	1.337	0.305	2.289	0.545	0.706
Nano Banana 2	0.723	0.387	0.994	0.444	0.832
FLUX.2 Max	0.761	0.040	0.741	0.546	0.714

Table 3: Post-result decomposition and scalar calibration. $D = D_{\text{aggregate}} + V_{\text{scene}}$, whereas $D = 1 - 2\beta + Q$ separates alignment and response magnitude. D_{held} evaluates a scalar fitted on the other 13 scenes. These are descriptive feature-space counterfactuals, without new significance tests or realizability claims.

Class-specific training targets give median .702 and range $[-.248, 1.946]$. These are finite-pool control distributions with different estimated normalizers, not confidence intervals or direct tests against the models. Their width prevents treating the model errors as calibrated degrees of artistic failure.

5.4 Common shifts and distributional proximity answer other questions

The common component accounts for 82.5–95.7% of global cross-repeat-corrected named/free squared change. Computing the same decomposition within scenes before averaging gives 88.6–97.2%. Thus global pooling is not responsible for the high common fraction in this experiment. Neither fraction is a percentage of images or of artistic style. Generic/common direction cosines range from .693 to .888; the generic shift is smaller for every model. The generic clause reproduces part of the common direction, without explaining all of it or removing possible shared non-additive transformations.

Naming lowers empirical reference energy relative to the generic clause in 21 of 24 model–painter cells. A correction respecting the two-repeat scene strata retains these 21 signs. The exceptions remain Flare for Monet and Pissarro, and Sunburst for Monet. Energy and primary contrast error give different point orderings: Sunburst has the lowest named energy for Sisley and Cézanne, whereas FLUX has the lowest for Monet and Pissarro. All 24 generated/reference variance ratios are below one (.248–.877), but these compare a fixed brief panel with broader historical collections. Complete four-painter tables and distributions appear in Appendix E.

FLUX's point estimate remains lowest after each scene deletion and under the square-window, reference-class, family-weight and covariance sensitivities. The original square and class-weighted analyses retain both resolved comparisons; the added weighting diagnostics introduce no new tests. Removing texture leaves Flare and Sunburst errors of 1.580 and 1.604. Persistence under these choices shows that a single texture family or broad content mixture does not explain all measured disagreement; it does not validate a unique artistic metric.

6 Discussion

The main finding is a separation between three claims: painter names induce reference-aligned variation; a configuration has lower error than another on a specified target; and that configuration faithfully reproduces an artist's style. The experiment supports the first claim for all six configurations and two comparisons under the second. It does not establish the third. Nor does it establish that any configuration improves on the no-artist-contrast benchmark in absolute squared error. Positive alignment and that absolute benchmark answer different questions.

The response-strength decomposition explains part of the model ordering. GPT Image 2 has a slope close to one, yet the total magnitude of its artist contrasts exceeds their reference-aligned component. FLUX has a weaker response and lower uncalibrated error. Scalar calibration on other scenes reduces error for every configuration and gives GPT Image 2 the smallest calibrated point estimate. This is evidence that response calibration influences the ranking, not that a feature-space rescaling is a realizable image intervention or that GPT Image 2 has the best artistic representation. The unresolved primary GPT Image 2–FLUX contrast remains unresolved.

Artist-pair analysis also changes the interpretation of positive aggregate alignment. The dominant reference contrast receives stronger responses than the remaining components in every model. In particular, positive overall slopes coexist with near-zero or reversed Monet–Sisley slopes in several configurations. A response aligned with a broad separation among the four painters is not equivalent to recovering all their distinctions.

A generic painting clause reproduces part of the common named shift; artist-contrast magnitude and scene dependence account algebraically for other components. These are observable response patterns, not identified internal mechanisms. Common non-additive transformations, artist-associated content substitution, and differences in reproduction processes remain alternatives to an incorrect artist representation. Training imbalance, preference optimization and guidance are possible upstream causes, but were not manipulated. The separate palette and fixed-map evidence in Appendix F, including the contrary prospective result, does not identify a universal mechanism.

6.1 What remains unvalidated

Genuine class-specific painting contrasts differ from the pooled target, and small disjoint real/reference panels yield widely varying corrected errors. The present collection cannot define a narrow artistic-fidelity threshold. Coarse within-class comparisons reduce one source of mismatch but do not match viewpoint, season, composition, career period or actual generated content. The rule-selected images expose concrete problems: the Sisley and Pissarro reference files include calibration strips retained by the primary full-frame measurement, and the Cézanne street scene is labeled water-organized from its title metadata (Appendix D). These examples establish that the reference target can contain non-painting content and imperfect subject labels; they do not estimate their prevalence or contribution to the reported errors. An output could also improve reference agreement by introducing artist-associated subjects that depart from the requested scene. The image panels permit inspection; they are not a blinded assessment of scene adherence or artist resemblance.

The 31 coordinates are transparent measurements, not a validated perceptual instrument. Equal family weighting and development-only covariance weighting retain FLUX's lowest primary point estimate, but cannot determine the right weighting of artistic properties. A previous processing challenge found that resizing and blur materially changed texture measurements; eight-neighbor local-binary-pattern entropy accounted for 83.9% of squared texture displacement under the tested blur, not of the present model error. Square windows and without-texture analyses do not control photographic illumination, sharpening, color correction or conservation state. Independent captures and style-oriented representations would provide different, still imperfect, validation routes.

The uncertainty analysis conditions on this reference panel and scaler. Synthetic checks illustrate acceptable family coverage under specified independent-error models and severe failure under shared-state errors; they do not establish the actual generation process. The 14 fixed briefs, two repeats, four related painters and one clause template restrict external inference. No independently collected replication, human style validation or source-held-out measurement validation has been completed. These gaps limit the artistic interpretation even when numerical replay succeeds.

7 Conclusion

Painter names produce detectable, reference-aligned digital contrasts beyond a common additive painting response. Their amplitude, scene dependence and agreement with pooled painting statistics nevertheless differ. FLUX has lower declared error than the two GPT Image 2.5 configurations, while held-scene scalar calibration and artist-pair analysis qualify any broader ranking. The contribution is a controlled demonstration that response detection, calibration and collection-relative agreement are separate evaluation questions. Establishing which disagreements represent artistic failure requires validation beyond the present feature-space target.

Data and code availability

The project repository contains protocols, exact prompts, request configurations, reference metadata, scalers, retained feature vectors and numerical replay code. The primary six-model results and the post-result diagnostics have separate records; the original inferential family is unchanged. Earlier experiments have versioned releases; the current addition has not yet been archived as an immutable public release.

Appendix D supplies 36 generated examples covering all model/clauses for one fixed brief, plus four reference reproductions. Stable image identifiers, exact prompts, source URLs, recorded rights and byte hashes accompany these panels. They support visual inspection but are not the full-resolution measurement inputs. Public numerical replay begins with retained vectors; it does not independently reproduce feature extraction or acquisition. Full-resolution artworks, generated images and response bodies remain outside the compact packages. Archiving permissible exact pixels and validating recovery from reference source records remain necessary for measurement-level reproducibility.

References

Andrea Asperti. On the separation of human and AI-generated images in CLIP embedding space. arXiv:2608.25609, version 1, 2026. URL https://arxiv.org/abs/2608.25609. Preprint.

Andrea Asperti, Franky George, Tiberio Marras, Razvan Ciprian Stricescu, and Fabio Zanotti. A critical assessment of modern generative models' ability to replicate artistic styles. *Big Data and Cognitive Computing*, 9(9):231, 2025. doi: 10.3390/bdcc9090231.

Yaniv Benny, Tomer Galanti, Sagie Benaim, and Lior Wolf. Evaluation metrics for conditional image generation. *International Journal of Computer Vision*, 129:1712–1731, 2021. doi: 10.1007/s11263-020-01424-w. URL https://link.springer.com/article/10.1007/s11263-020-01424-w.

Adrien Deliège, Jeanne Marlot, Marc Van Droogenbroeck, and Maria Giulia Dondero. How good is the machine at the imitation game? on stylistic characteristics of AI-generated images. *Journal of Imaging*, 11(12):429, 2025. doi: 10.3390/jimaging11120429. URL https://pmc.ncbi.nlm.nih.gov/articles/PMC12734345/.

Jörn Diedrichsen and Nikolaus Kriegeskorte. Representational models: A common framework for understanding encoding, pattern-component, and representational-similarity analysis. *PLOS Computational Biology*, 13(4):e1005508, 2017. doi: 10.1371/journal.pcbi.1005508.

Tarian Fu, Javier Conde, Gonzalo Martínez, Pedro Reviriego, Elena Merino-Gómez, and Fernando Moral. Artificial intelligence and misinformation in art: Can vision language models judge the hand or the machine behind the canvas? arXiv:2508.01408, version 1, 2025. URL https://arxiv.org/abs/2508.01408v1.

D. Kim, S.-W. Son, and H. Jeong. Large-scale quantitative analysis of painting arts. *Scientific Reports*, 4:7370, 2014. doi: 10.1038/srep07370.

J. Kim, B. Lee, T. You, and J. Yun. Context-aware multimodal AI navigates hidden pathways in five centuries of art evolution. *Proceedings of the National Academy of Sciences*, 123(30): e2517969123, 2026. doi: 10.1073/pnas.2517969123.

B. Lee, M. K. Seo, D. Kim, I.-S. Shin, M. Schich, H. Jeong, and S. K. Han. Dissecting landscape art history with information theory. *Proceedings of the National Academy of Sciences*, 117 (43):26580–26590, 2020. doi: 10.1073/pnas.2011927117.

Stéphane G. Mallat. A theory for multiresolution signal decomposition: The wavelet representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(7):674–693, 1989. doi: 10.1109/34.192463. URL https://www.di.ens.fr/~mallat/papiers/MallatTheory89.pdf.

Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, pages 7176–7185, 2020. URL https://proceedings.mlr.press/v119/naeem20a.html.

Timo Ojala, Matti Pietikäinen, and Topi Mäenpää. Multiresolution gray-scale and rotation invariant texture classification with local binary patterns. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(7):971–987, 2002. doi: 10.1109/TPAMI.2002.1017623.

Gaurav Parmar, Richard Zhang, and Jun-Yan Zhu. On aliased resizing and surprising subtleties in GAN evaluation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://arxiv.org/abs/2104.11222.

Gaurav Sharma, Wencheng Wu, and Edul N. Dalal. The CIEDE2000 color-difference formula: Implementation notes, supplementary test data, and mathematical observations. *Color Research and Application*, 30(1):21–30, 2005. doi: 10.1002/col.20070.

H. Y. D. Sigaki, M. Perc, and H. V. Ribeiro. History of art paintings through the lens of entropy and complexity. *Proceedings of the National Academy of Sciences*, 115(37):E8585–E8594, 2018. doi: 10.1073/pnas.1800083115.

Gowthami Somepalli, Anubhav Gupta, Kamal Gupta, Shramay Palta, Micah Goldblum, Jonas Geiping, Abhinav Shrivastava, and Tom Goldstein. Investigating style similarity in diffusion models. In *European Conference on Computer Vision*, 2024. URL https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/08294.pdf.

Grace Su, Sheng-Yu Wang, Aaron Hertzmann, Eli Shechtman, Jun-Yan Zhu, and Richard Zhang. Identifying prompted artist names from generated images, 2025. URL https://arxiv.org/abs/2507.18633v1. Version 1.

Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013. doi: 10.1016/j.jspi.2013.03.018.

Denny Vrandečić and Markus Krötzsch. Wikidata: A free collaborative knowledgebase. *Communications of the ACM*, 57(10):78–85, 2014. doi: 10.1145/2629489.

Wikimedia Commons. Commons:GLAM, 2026. URL https://commons.wikimedia.org/wiki/Commons:GLAM. Accessed 11 September 2026.

A Feature definitions

Color coordinates use CIELAB under D65 and the 2° observer. Lightness and chroma each contribute a median and IQR. Chromatic fraction is the share of pixels with $C^* \geq 5$. Hue concentration is the magnitude of the chroma-weighted circular mean; hue entropy uses 24 equally spaced bins and is normalized by their log count. Both use chroma weights and are zero when fewer than 1% of pixels are chromatic. At each of three lags (1%, 4% and 16% of the short side), the median is taken over pooled horizontal and vertical CIEDE2000 differences. A log–log slope summarizes their scale dependence (Sharma et al., 2005).

Spatial and texture coordinates use linear-sRGB luminance. A Tukey window (parameter .1) precedes Fourier analysis. Thirty-two logarithmic radial bins cover 4–128 cycles per short side; a Theil–Sen line gives spectral slope and its root-mean-square residual. Power-weighted axial concentration measures anisotropy. Scharr derivatives supply gradient median and IQR, an 18-bin magnitude-weighted orientation entropy, and horizontal–vertical balance. Mean Jensen–Shannon divergence of quadrant orientation histograms from the global histogram measures regional variation.

Four log mean-squared detail energies come from a four-level undecimated `db2` wavelet transform, followed by their linear slope and mean second difference (Mallat, 1989). Uniform local-binary-pattern entropy uses $(P, R) = (8, 1), (16, 2), (32, 4)$ on quantized luminance (Ojala et al., 2002). Three median local coefficients of variation use window widths obtained by rounding 1%, 4% and 16% of the short side and increasing even widths to the next odd integer, with a 10^{-6} mean floor. The fixed implementation records boundary handling and stabilizing constants. The separate scaler contains 101 Monet, 36 Sisley, 48 Pissarro and 36 Cézanne works. Equal painter mass is used to calculate each weighted median and IQR.

B Estimator details and sensitivity analyses

Let $\theta_{msa} = \mathbb{E}[d_{msak}]$ and $d_{msak} = \theta_{msa} + \eta_{msak}$. If repeat errors have zero mean and $\mathbb{E}\langle \eta_{msa1}, \eta_{msa2} \rangle = 0$, then

$$\mathbb{E}[\widehat{D}_{ms}] = H^{-1} \sum_{a} \|\theta_{msa} - r_a\|^2.$$

Centering induces dependence between artists within a repeat, but independent repeat blocks preserve the vanishing cross-repeat term. A common response has $\theta_{msa} = 0$ and hence expected $D = 1$, even if it shifts the entire named collection toward the references. A correctly aligned but rescaled response $\theta_{msa} = br_a$ gives expected $D = (b - 1)^2$. Orthogonal deviations add their squared magnitude. This identity explains why stronger painter contrasts need not have lower reference-relative squared error.

For each model, global distortion applies the same cross-product estimator to the scene-averaged contrasts. Its target differs from the mean scene-wise error through the variation term in Equation 5; this term is not necessarily a failure. Both are reported, without selecting the smaller as the primary result. The primary comparison weights scenes and artists equally, irrespective of the reference count. The report also retains each painter's additive contribution to D. These sum to the model error but are not standalone artist-fidelity scores, because the centering in Equation 1 couples the four named conditions.

For a model contrast, compute one error difference per complete paired scene. Its standard error is the sample standard deviation of those differences divided by $\sqrt{n}$. The simultaneous half-width is $t_{n-1,1-.05/(2\cdot 21)}$ SE. Six slope intervals use the same family adjustment. Additional nominal intervals and 5,000 paired-scene bootstrap draws are descriptive checks, not alternative routes to significance. Reference sensitivity draws 1,000 bootstrap samples independently within each painter, recomputes reference means and H, and holds generated vectors and scaling fixed. It describes reference-panel dependence rather than all uncertainty jointly.

The four-class reference sensitivity uses a fixed lexicon applied to source titles: water-organized, built-place-organized, route-organized, and open or wooded land. Each class mean receives one-quarter weight within a painter. Resampling within artist/class strata conditions on those labels and does not account for classification error. The older controlled naming panel used a separate three-class visual coding scheme, described below.

The shared-effect diagnostic first averages each arm over scenes within repeat. Let h_{ak} be the named-minus-free difference and $c_k = \frac{1}{4}\sum_a h_{ak}$. The cross-repeat common and artist-specific squared magnitudes are $4\langle c_1, c_2\rangle$ and $\sum_a \langle h_{a1} - c_1, h_{a2} - c_2\rangle$. The common fraction divides the first magnitude by their sum and is reported only when that sum is positive; finite corrected components can be negative, so the ratio is not a probability. Generic/common cosine uses their repeat-averaged vectors. This descriptive global statistic is separate from the conditional recovery endpoint.

C Post-result target diagnostics

These analyses use the completed 1,008 generated vectors, existing references and unchanged scaler. Their choices were recorded after the primary results and reviewer comments were known. They are not new confirmatory experiments. The accompanying result contains every scalar fit, artist-pair result, artist omission, label assignment, coordinate contribution and control draw.

C.1 Content-specific targets and genuine-painting controls

The recorded reference classes are water-organized, built-place-organized, route-organized, and open or wooded land, assigned by a fixed source-title lexicon. Counts in that order are $(191, 67, 8, 31)$ for Monet, $(52, 29, 9, 16)$ for Sisley, $(28, 77, 12, 24)$ for Pissarro and $(31, 28, 12, 34)$ for Cézanne. Class-specific means are centered across painters just as in the pooled target. Their mean squared departure from the pooled target is $.438H$, a description of observed reference means that includes finite-sample error.

For the generated comparison, the four water, four built and three land briefs use the corresponding class means; the three mixed briefs are excluded. We divide the summed class-specific cross-product error by $H_c = S^{-1}\sum_s \sum_a \|r_{a,c(s)}\|^2 = 7.413$, compared with pooled $H = 5.915$. Each column of Table 4 therefore has its own denominator. The unchanged 11-scene pooled comparison isolates the scene-subset choice. These title-derived strata have 16–191 reference works per artist/class among the three included classes, but do not supply visually matched historical scenes.

For each of 1,000 genuine-painting controls, works in each artist/class cell are randomly divided into disjoint halves, with the smaller half used for target estimation. A pooled control samples two independent works with replacement from each painter's held-out half for each of 14 pseudo-scenes. A class-stratified control samples within the held-out class for each of the 11 water, built or land pseudo-scenes. It is compared with both pooled and class-specific means from the training halves, using each target's own squared magnitude. The random seed is 2026091201. Sampling can repeat a held-out work, but no held-out work is used to estimate that split's target. The development scaler is fixed throughout.

These controls reveal finite-reference instability and content dependence; they do not establish a universal real-art baseline. Training-target estimation, held-out mean differences and two-draw noise all contribute to their spread. Stratifying on title classes does not hold out capture sources or remove classification error. In particular, the class-specific control's median error .702 and central interval $[-.248, 1.946]$ do not justify a perceptual acceptance threshold or an assertion that model errors above one indicate artistic failure.

Model	11 common briefs Pooled target	Class target	14 briefs Equal family	Covariance
GPT Image 1	1.557	1.366	1.551	2.103
GPT Image 2	1.099	1.130	1.243	1.511
2.5 Flare	1.726	1.539	1.765	1.636
2.5 Sunburst	1.633	1.525	1.680	1.617
Nano Banana 2	0.987	0.948	1.203	1.385
FLUX.2 Max	0.750	0.786	0.808	0.928

Table 4: Descriptive error under alternative targets and metrics. The class target uses title-derived water, built and land strata; the three mixed briefs are excluded. Each column uses its own reference normalizer; absolute values across columns are not directly comparable. Covariance is fitted only on development works, with fixed 50% shrinkage.

Model	Reference-component slopes First	Second	Third	Slope after omitting Monet	Sisley	Pissarro	Cézanne
GPT Image 1	1.248	0.310	0.356	1.140	1.072	0.840	0.389
GPT Image 2	1.156	0.617	0.800	1.195	0.972	0.992	0.651
2.5 Flare	1.022	0.211	0.383	1.034	0.734	0.802	0.222
2.5 Sunburst	1.082	0.250	0.417	1.116	0.840	0.762	0.284
Nano Banana 2	0.513	0.371	0.195	0.569	0.446	0.388	0.276
FLUX.2 Max	0.666	0.021	0.183	0.539	0.511	0.527	0.096

Table 5: Artist coverage of the aligned response. Reference components account for 66.3%, 20.6% and 13.0% of centroid variation. Each omission recomputes the target and recenters the other three artists. These slopes are descriptive, not artist-level significance tests.

C.2 Magnitude, artist coverage and coordinate weighting

The cross-repeat version of Equation 5 is exact for the retained vectors. Its variation term replaces squared deviations with cross-products between scene-centered repeats. The term can be negative in finite samples, although its independent-repeat population target is nonnegative. Scalar calibration fits only the other 13 scenes. All fold denominators Q are positive. GPT Image 2's fitted scalars range from .438 to .459 and FLUX's from .598 to .694; all fold-specific scores are retained. No calibration parameter is selected by minimizing its own held-out score.

For artist pairs, both generated and reference contrasts are direct differences between the two artists; the denominator is the squared reference-pair distance. Leaving one artist out recenters the remaining three generated and reference conditions. Reference singular components use the three rank-one terms of the reference-mean matrix, and project onto each such term separately. Table 5 shows that aggregate positive alignment is uneven across those components. The 24 permutations rearrange generated artist labels while leaving the reference fixed. Their ranks are not exact permutation p-values: the named conditions are not exchangeable under an established null.

Equal-family weighting gives each coordinate a weight inversely proportional to its family's size (11, 8 or 12). The covariance sensitivity uses the 221 development works with equal total mass per painter. For their weighted covariance Σ, the metric matrix is $[.5\Sigma + .5\,\mathrm{tr}(\Sigma)I/31]^{-1}$; the shrinkage is fixed, without tuning on the evaluated images. These changes address correlated coordinate weighting but are not independently validated artistic metrics. FLUX retains the smallest point estimate under each of 31 single-coordinate deletions. The largest individual primary-error contribution varies by model: 32-neighbor LBP entropy for GPT Image 1, spectral residual for GPT Image 2, wavelet curvature for the two 2.5 variants, and quadrant orientation divergence for Nano Banana 2 and FLUX. All contributions, including negative terms, are reported rather than selecting a favorable coordinate subset.

C.3 A design-aware energy correction

The primary energy values use empirical distributions, including their zero diagonal distances. To estimate energy between a fixed empirical reference and the equally weighted mixture of scene-conditional output distributions, cross-scene generated pairs already have the required independent draws. Only the same-scene generated term needs adjustment. With two repeats and S scenes, the descriptive correction is

$$\widehat{\mathcal{E}}_{\text{strata}} = \widehat{\mathcal{E}}_{\text{empirical}} - \frac{1}{2S^2} \sum_{s} \|y_{s1} - y_{s2}\|.$$

The reference within-distance remains empirical because this target conditions on the finite reference panel. A pooled IID $n/(n-1)$ multiplier would also alter cross-scene terms and estimate a different target. This correction retains the 21 negative named-minus-generic differences. It does not solve unequal historical and generated content mixtures, and no new energy tests are performed.

C.4 Uncertainty and synthetic coverage checks

The Student procedure uses 14 scene summaries. For independent scene errors with fixed conditional means δ_s, its estimated variance has expectation

$$\mathbb{E}[s_{\delta}^2/S] = \operatorname{Var}(\bar{\hat{\delta}}) + \frac{1}{S(S-1)} \sum_{s} (\delta_s - \bar{\delta})^2.$$

It therefore includes real between-scene heterogeneity as well as request noise. This explains why it is not a direct estimate of fresh-request variance for exactly the fixed panel. It also does not provide an exact coverage guarantee: non-Gaussian cross-products and dependence can affect the Student approximation.

We simulate 5,000 repetitions per specified scenario (seed 2026091202), with six configurations, 14 scenes, two repeats and the same 21-endpoint family. The reference configuration is normalized to unit squared magnitude. Synthetic mean contrasts combine fixed reference slopes with orthogonal components of magnitudes similar to the reported estimates; random directions are artist-centered. Independent centered noise has total expected squared magnitude .5. The scenarios use Gaussian errors; variance-standardized Student t_3 errors with scene-specific scale factors from .5 to 2; Gaussian errors with additional fixed artist-by-scene interactions; and a shared-state violation. The last allocates half the error variance to a model-specific random component shared by both repeats and every scene in a simulated run. These are explicit stress models, not estimates of the actual service process.

Simultaneous coverage of all 21 known fixed-panel quantities is 95.96%, 97.42% and 99.68% in the first three scenarios. It falls to .04% in the shared-state scenario, where mean corrected-error bias is approximately .25, as expected from the retained cross-repeat covariance. Monte Carlo standard error for a coverage probability is $\sqrt{p(1-p)/5000}$. The simulations show how the procedure can succeed or fail under specified assumptions; they cannot verify actual repeat independence or calibrate the original intervals from two observations per cell. No revised significance threshold is selected from them.

D Inspectable image examples

The generated panel uses the first declared brief and first repeat for every model and clause, selected by this fixed rule after numerical analysis. The brief reads: "A broad river bends past a gravel bank. Three tall trees stand on the far bank, with a low hill behind them under a pale overcast sky." Every prompt ends with "No text or frame." Artist-free, generic and named clauses otherwise follow Section 3. Images were not selected by visual success, distance or agreement with the paper's claims.

Reference examples are the lexicographically first measured water-class work identifier for each painter with recorded public-domain status. They are examples from the comparison collection, not matched depictions of the generated brief. The Sisley and Pissarro examples visibly retain calibration strips. Their primary measurement records have zero crop fraction, so those non-painting regions were not masked. The Cézanne example, *Crossroad of Rue Rémy, Auvers*, depicts a street despite its title-derived water class. These observations concern four selected examples, not a prevalence audit. They are retained to expose the source and labeling problems, without recropping, reclassifying or recomputing the sealed analysis. The accompanying inspection manifest identifies exact source bytes, work metadata and generated requests. These reduced display panels permit qualitative inspection of content and rendering, not independent feature extraction, blinded scene-adherence scoring or artist-fidelity validation.

Figure 3: Rule-selected reference examples. Columns identify painters and recorded Wikidata work identifiers. Calibration strips are visible beside the Sisley and Pissarro reproductions; the Cézanne example reveals a title-class mismatch. Source URLs and public-domain metadata are retained with the inspection manifest. The works are not exact content matches to the common generated brief.

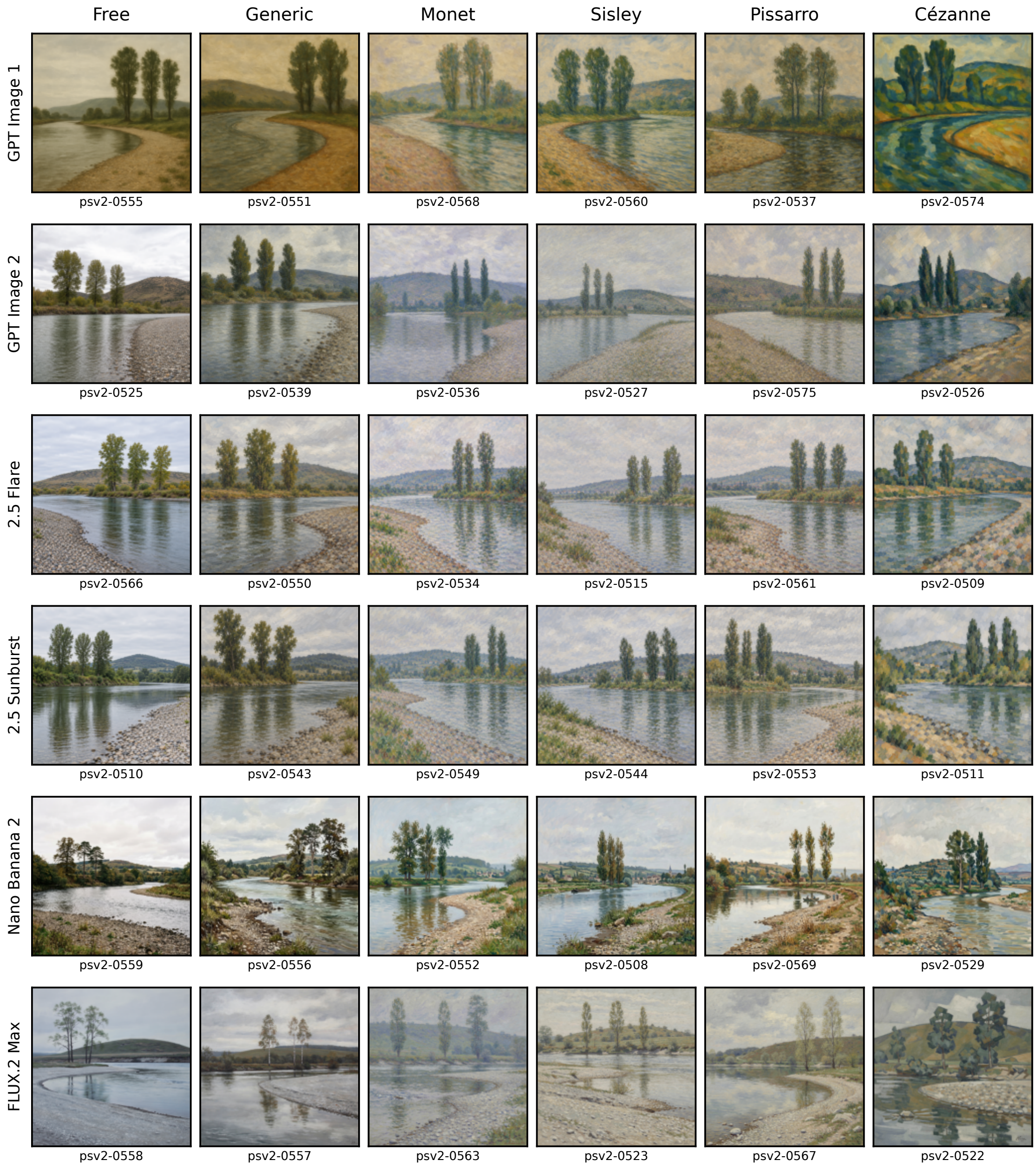

Figure 4: One common brief across all six models and clauses, first repeat. Every thumbnail carries its stable request identifier. Rows compare recorded configurations; columns compare clauses. Selection is deterministic and post-result, without appearance-based curation. No human evaluation is inferred from this panel.

E Complete model and painter summaries

Table 6 reports the full model-comparison family; Table 7 retains distributional outcomes for every painter and model. The latter do not inherit the primary model-contrast significance tests. Their energy values concern empirical distributions of 28 named or generic outputs against the recorded reference panel.

Model A	Model B	$D_A - D_B$ [simultaneous interval]
GPT Image 1	GPT Image 2	0.346 $[-0.400, 1.093]$
GPT Image 1	2.5 Flare	-0.165 $[-1.027, 0.697]$
GPT Image 1	2.5 Sunburst	-0.069 $[-1.181, 1.043]$
GPT Image 1	Nano Banana 2	0.463 $[-0.602, 1.528]$
GPT Image 1	FLUX.2 Max	0.771 $[-0.037, 1.580]$
GPT Image 2	2.5 Flare	-0.512 $[-1.150, 0.126]$
GPT Image 2	2.5 Sunburst	-0.415 $[-1.181, 0.350]$
GPT Image 2	Nano Banana 2	0.116 $[-0.922, 1.155]$
GPT Image 2	FLUX.2 Max	0.425 $[-0.161, 1.011]$
2.5 Flare	2.5 Sunburst	0.096 $[-0.286, 0.478]$
2.5 Flare	Nano Banana 2	0.628 $[-0.283, 1.540]$
2.5 Flare	FLUX.2 Max	0.937 $[0.256, 1.618]$
2.5 Sunburst	Nano Banana 2	0.532 $[-0.436, 1.500]$
2.5 Sunburst	FLUX.2 Max	0.840 $[0.058, 1.623]$
Nano Banana 2	FLUX.2 Max	0.309 $[-0.847, 1.464]$

Table 6: All 15 paired model comparisons in the prespecified inferential family. A negative contrast favors model A on the stated feature-geometry target. Flare and Sunburst denote the corresponding GPT Image 2.5 models.

Model	Monet	Sisley	Pissarro	Cézanne
GPT Image 1	−2.515 / 0.479	−2.739 / 0.694	−2.628 / 0.603	−1.336 / 0.598
GPT Image 2	−0.602 / 0.283	−1.092 / 0.312	−0.195 / 0.303	−0.373 / 0.303
2.5 Flare	0.962 / 0.269	−0.008 / 0.371	0.429 / 0.321	−1.013 / 0.275
2.5 Sunburst	0.267 / 0.248	−0.718 / 0.274	−0.091 / 0.273	−1.653 / 0.318
Nano Banana 2	−0.513 / 0.829	−1.381 / 0.708	−0.741 / 0.572	−1.102 / 0.877
FLUX.2 Max	−0.707 / 0.340	−0.886 / 0.416	−1.080 / 0.570	−1.431 / 0.622

Table 7: Complementary outcomes for all four painters: each cell gives named-minus-generic reference energy / generated-to-reference total variance. These are descriptive empirical distribution summaries, without additional tests. Lower energy change favors naming over the generic clause; a variance ratio below one indicates a narrower generated feature distribution.

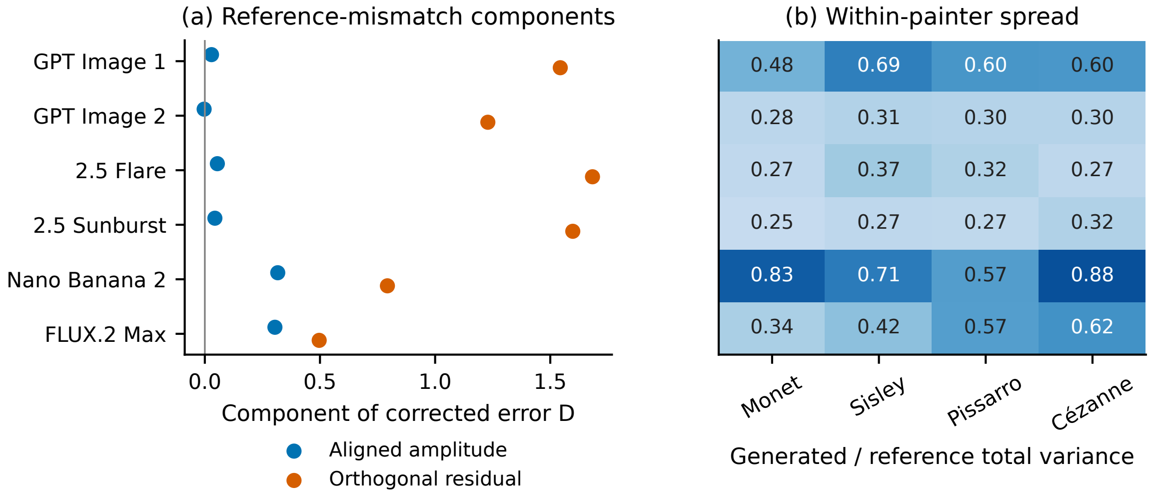

Figure 5: Complementary descriptive summaries. (a) Reference-aligned amplitude error and the residual orthogonal to the stacked reference configuration. (b) Generated/reference variance ratios for every model and painter. Total spread includes variation across fixed briefs and is not a mode-collapse measure.

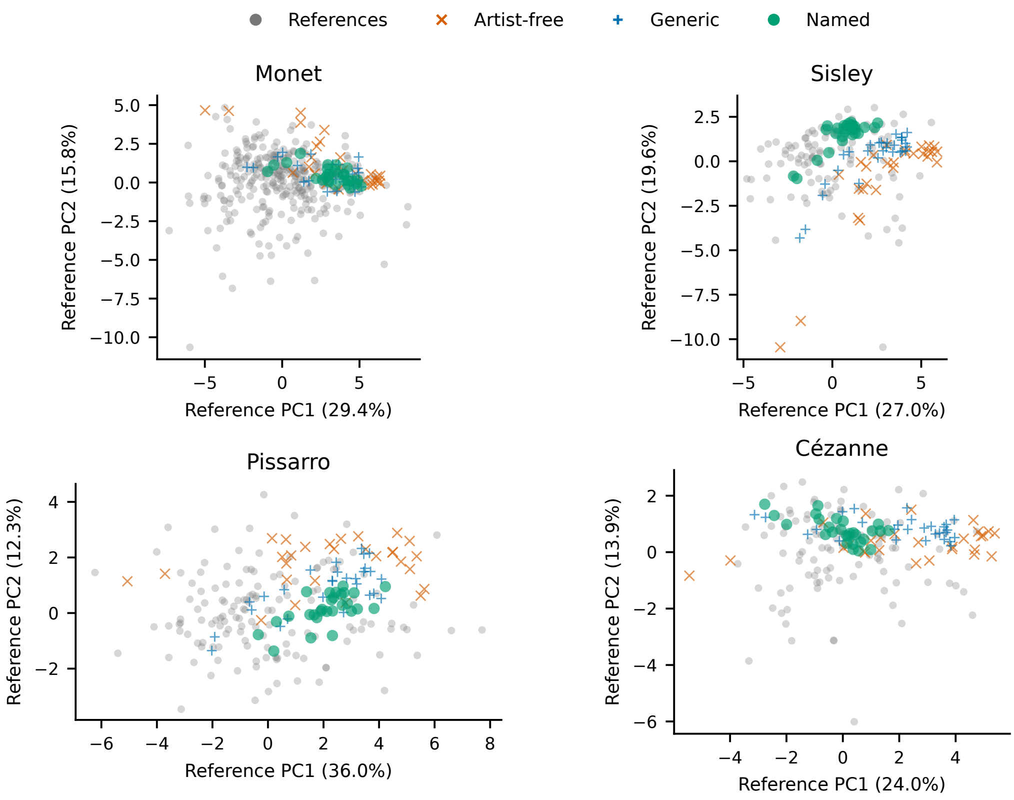

Figure 6: All four painters' reference and generated distributions, illustrated with GPT Image 2.5 Sunburst. This model illustration was selected before the primary feature outcomes were inspected. Each panel uses principal components fitted only to that painter's reference works, so axes differ between painters. The 14 generated briefs and two repeats have a different content distribution from the historical panel. Projections are descriptive; numerical summaries use all 31 coordinates.

F Supporting interventions

These earlier cohorts address related but different questions (Table 8). They are not pooled with the six-model experiment or treated as independent-investigator replications.

Cohort	Comparison	Role in this paper
Four-painter exploration	649 references; 1,920 generated outputs	Describes reference/generated separation and motivates the controlled test.
Controlled naming	1,006 outputs; 38 Monet and 32 Cézanne references	Six named/free and two detailed/short prompt contrasts; subsequent transformation and conditional-variance diagnostics.
Palette interventions	Two separate 192-output collections	Named-minus-generic chroma response; no pooling across collections.
Generic-clause comparison	96 Cézanne/generic outputs on 24 new scenes	Tests a proximity gain beyond a generic painting clause.
Fixed-map transfer	240 FLUX/Cézanne outputs on 12 new scenes	Prospective counterexample to the historical opposing energy/prediction ordering.

Table 8: Supporting evidence is kept separate from the new six-model experiment. Different prompts, panels and finite-sample targets do not form one pooled test.

F.1 Proximity and spread under naming

The four-painter exploration used 1,536 named and 384 artist-free outputs, three prompt methods and 16 scenes. Its 24 model–painter–method cells had trace ratios .206–.376 and grouped generated/reference discrimination accuracy .940–.983. Pooled named energy medians for Monet, Sisley, Pissarro and Cézanne were 2.068, 1.678, 1.975 and 2.142; corresponding artist-free medians were 1.732, 1.878, 1.854 and 1.657. Only Sisley had a lower pooled named median. Matched-size real/real medians were .193, .215, .164 and .199. These post-result summaries describe that exploration, not a general naming effect.

The later controlled panel selected Monet and Cézanne after exploration. It crossed 24 scenes with three models, two clauses and three repeats; 142 short-scene outputs supplied two prompt-length comparisons. Reference panels contained 38 Monet and 32 Cézanne works, with water/built/land masses based on maintainer-run LLM visual coding. Named-minus-free energy changes were $-.846, -1.818$ for Nano Banana 2, $-.625, -.896$ for FLUX, and $-.234, -.056$ for GPT Image 2 (Monet, Cézanne). The first four rejected in the original family of eight. A separate 72-output FLUX collection repeated both directions $(-.695, -.952)$ with rejection in its own four-endpoint family.

A standalone 96-output Cézanne comparison on 24 new scenes reduced energy relative to the generic clause by 1.195 ($p = .00001$, fixed $\alpha = .025$), with named/generic trace ratio .552. It randomized clauses within 48 two-position blocks, used reference class masses $(3, 11, 18)/32$, and tested a sharp no-effect null with 99,999 within-block sign draws and plus-one correction. This tests clause effects, not equality of energy alone. A generic clause therefore did not reproduce the full proximity gain in that panel, but proximity does not establish correct four-artist contrasts. Likewise, lower total spread does not establish loss of scene information: controlled FLUX/Monet held-repetition scene retrieval rose from 44.4% to 59.7% under naming. This measures relative feature distinguishability, not verified prompt adherence.

F.2 Palette responsiveness and a contrary transfer result

Two separate 192-output palette collections used six scenes, four repeats, muted/vivid instructions and free, generic, Monet and Cézanne clauses. The primary name-minus-generic chroma-response interactions were unresolved: $-.284, -.018$ in the first collection and $-.159, .037$ in the later one. Their respective adjusted intervals spanned zero; the original family had two interactions and the later family included two additional FLUX endpoints. Secondary generic-minus-free responses were $-.853$ and $-.891$, with nominal 95% intervals excluding zero. Thus generic painting can attenuate measured palette response without an identified additional name effect. Two palette levels do not distinguish reduced gain from saturation or an operating-range shift, and unresolved interactions do not demonstrate equivalence.

Historical generated-only moment maps were fitted on training scenes and assessed on held-out scenes. Actual naming had lower reference energy than translation plus isotropic scaling in all six model–painter averages. Adding scale improved named-mean prediction in all six while worsening reference energy in five. That ordering did not transfer to a prospective 240-output FLUX/Cézanne comparison with 12 new scenes and ten repeats, using the original fits unchanged. Translation/scaling improved both energy and repeat-corrected prediction error relative to translation: $-.390$ $[-.509, -.271]$ and -5.047 $[-7.880, -2.214]$, with approximate simultaneous jackknife intervals. The achieved prediction-error half-width exceeded its planning criterion. This contrary result is retained without extra sampling; it rules out treating the historical opposing ordering as a universal mechanism. These fitted named/free maps are distinct from the review-driven artist-contrast scalar counterfactual in Table 3.

G Collection provenance and computational scope

Historical GPT Image 1 and GPT Image 2 labels identify the requested model names. A later technical negative control showed that the earlier image interface also accepted an invented image-model identifier. This does not show that valid names select the same underlying model, but successful request acceptance alone cannot authenticate their separation. Historical contrasts are therefore interpreted as recorded request-group comparisons. The current experiment uses explicit documented model identifiers and fixed routes; it does not reuse historical images as verified examples of the new variants. All current outputs decode to the requested square dimensions, but their response bodies do not echo model or quality fields. Labels therefore denote the documented request configurations, without independent checkpoint attestation. Underlying checkpoints and training corpora are not available for inspection.

The present experiment excludes technical probes and an earlier attempt closed before feature measurement because model selection could not be qualified. No image is shared between these collections. The earlier incomplete generic-clause cohort is likewise not pooled with its complete 96-image successor. Exact assignments, attempt outcomes and measurement exclusions are retained in the versioned manifests; selection was not based on visual success.